Roll No. | | | | | | | |

Paper Code:TBT-101

(B. Tech Biotechnology)

End Semester Examination, December 2025

I *Semester*

Paper Name: FUNDAMENTALS OF MEDICAL MICROBIOLOGY

Time: 3:00 Hours MM: 100

Note:

(i) All questions are compulsory.

(ii) Answer any TWO sub questions among a, b & c in each main question.

(iii) Total marks for each main question are twenty.

Q1	(20marks)	
(a)	Define infection and disease with suitable examples. Describe the major modes of transmission of infectious agents and explain how vectors such as mosquitoes and ticks facilitate their spread.	
(b)	Describe the composition of the natural skin microbiota and explain how these microorganisms, along with the skin's physical and chemical properties, help prevent pathogen entry.	**CO1**
(c)	Explain the morphological characteristics and infection sources of *Clostridium tetani*. How are the different clinical forms of tetanus distinguished from one another?	
Q2	(20 marks)	
(a)	Describe the general morphology of fungal pathogens responsible for causing mycoses. How do yeasts, molds, and dimorphic fungi can be isolated and cultivated in laboratory?	
(b)	Describe the morphology of *Rhabdovirus*. With the help of a labeled diagram, explain the pathogenesis of rabies infection and outline the laboratory methods used for its diagnosis.	**CO2**
(c)	Explain the morphology of the Human Immunodeficiency Virus (HIV). Describing the principle of ELISA, explain the mechanism of detecting the HIV p24 antigen in a patient's serum sample using a sandwich ELISA involving a chromogenic substrate.	
Q3	(20 marks)	
(a)	Explain the principle and application of the *agglutination inhibition test* in diagnosing the presence of the drug cocaine in a victim's biological sample.	
(b)	Describe the characteristics of effective immunogens that trigger an active immune response. Elaborate on the sequential mechanisms through which T cells are activated and mediate the immune defense against bacterial infections.	**CO3**
(c)	Describe the structure of an immunoglobulin and their different classes. How does its structure relate to its function?	
Q4	(20 marks)	
(a)	Elaborate on different types of clinical specimens commonly collected in a diagnosis of diseases? Elaborate on the composition, principle, and diagnostic role of Blood Agar in isolating and identifying *Streptococcus* species from patient-sputum specimens	
(b)	Discuss different classes of antibiotics and their mode of action? How does antibiotic disc diffusion assay is employed to evaluate bacterial sensitivity and guide the rational choice of antibiotic therapy.	**CO4**
(c)	Describe the National Programme for the Control of Vector-Borne Diseases. Discuss the various quality control measures employed in diagnostic microbiology laboratories	

Q5	(20 marks)	
(a)	How has biotechnology contributed to the development of different generations of vaccines? Discuss the major types of vaccines and their role in the prevention and treatment of human diseases.	
(b)	A pharmaceutical company plans to produce recombinant human insulin using genetic engineering. Describe the major steps and biomedical tools involved in developing recombinant insulin and explain its importance in diabetes management.	**CO5**
(c)	A biotechnology company aims to develop monoclonal antibodies against the Hepatitis B surface (HBs) antigen using hybridoma technology. Describe the process involved and explain how the selective " HAT" medium exploits the de novo and salvage pathways for selecting hybrid cells.	